The Generalized Local Rule and the Burge–RSK Involution

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Abstract

We analyse whether the Schützenberger involution relating the Burge and RSK local rules extends to the Q-enriched setting of [1]. The conclusion is negative: the Q-kernel $\Phi_{a,b}(t)$ is context-sensitive (it depends on the strip sizes at each face), and the Schützenberger involution changes the interior partition shapes, propagating Q-weight changes through the growth diagram. The failure is structurally informative: it shows that effective involutions for Q-positivity must be *local* (operating one face at a time), matching the recursive structure of the shifted factorisation $\Phi_{a,b}(t) = Q_a Q_b + t\,\Phi_{a+1,b+1}(t)$.

Contents

1 Setup **2**

2 Defining generalised RSK **3**

3 The Schützenberger involution **3**

4 The propagation obstruction **3**
4.1 First row . . . 3
4.2 Second row and beyond . . . 3

5 The $d = 1$ exception **4**

6 Context-sensitivity and global coupling **4**

7 What could work instead **4**
7.1 Peel/glue at the ALCD level . . . 4
7.2 The Q-kernel recurrence . . . 5
7.3 The commutator approach . . . 5

8 The structural lesson **5**

9 Open question **6**

10 The pair-based Q-local rule **6**
10.1 Why not diamonds . . . 6
10.2 The underlying object: pairs of ground-set elements . . . 6
10.3 The sign structure . . . 7
10.4 The cascade as coupling . . . 7

11 The involution programme on pairs **8**
11.1 The full signed sum 8
11.2 Factored vs. coupled involutions 8
11.3 Coupled involutions via cascade shift 9
11.4 The programme 9

12 The Q-Burge and Q-RSK rules **10**
12.1 Peel and glue on ground-set elements 10
12.2 The Q-Burge rule on pairs 10
12.3 The Q-RSK rule on pairs 11

13 The paired involution **11**
13.1 The Franklin-style comparison 11
13.2 Sign-reversal and weight preservation 12
13.3 Fixed-point structure 12
13.4 Why two rules are needed 12
13.5 Connection to the shifted factorisation 13

1 Setup

Recall the two parallel frameworks.

Algebraic. The fundamental commutation relation (cf. [3], [1]):

$$T(v)\,P(w) = \frac{1}{1-vw}\;P(w)\,T(v), \tag{1}$$

which at the mode level gives the cascade relation:

$$T_n\,P_m = \sum_{k=0}^{\min(n,m)} P_{m-k}\,T_{n-k}. \tag{2}$$

Bijective. The Burge local rule $\mathfrak{U}_{\alpha,\beta}$ has type signature

$$\mathfrak{U}_{\alpha,\beta}\colon \mathbb{Z}_{\geq 0} \times (D(\alpha) \cap D(\beta)) \to U(\alpha) \cap U(\beta), \tag{3}$$

where $U(\mu)$ (resp. $D(\lambda)$) is the set of partitions obtained by adding (resp. removing) a horizontal strip. The weight condition is:

$$|\lambda| = |\alpha| + |\beta| - |\nu| + k. \tag{4}$$

Each commutation step corresponds to one local rule application. The scalar factor $1/(1-vw)$ generates the face labels. Both the Burge and RSK local rules share the same type signature (3) and satisfy the same weight condition (4).

The Q-enriched setting. In the Q-enriched framework ([1], Part III), the face label k carries weight $Q_{a+k}(q)\cdot Q_{b+k}(q)$, where $a = |\alpha| - |\nu|$ and $b = |\beta| - |\nu|$ are the horizontal strip sizes entering the diamond. The face-label generating function becomes the Q-kernel:

$$\Phi_{a,b}(t) = \sum_{k\geq 0} Q_{a+k}\,Q_{b+k}\,t^k, \tag{5}$$

replacing the universal $1/(1-t)$ of the standard rule.

2 Defining generalised RSK

This works without difficulty. The RSK local rule has the same type signature as Burge:

$$\mathfrak{U}^{\mathrm{RSK}}_{\alpha,\beta}\colon \mathbb{Z}_{\geq 0} \times (D(\alpha) \cap D(\beta)) \to U(\alpha) \cap U(\beta), \tag{6}$$

with the same weight condition $|\lambda| = |\alpha| + |\beta| - |\nu| + k$.

Definition 1 (Q-enriched RSK)**.** The *Q-enriched RSK local rule* is defined identically to the Q-enriched Burge rule: same partition geometry, same cascade parameter k, face label k carries weight $Q_{a+k}(q) \cdot Q_{b+k}(q)$.

Since the inputs (ν, α, β, k) are the same for both rules, and the strip sizes a, b depend only on the input, the Q-weight at each individual face is the same for both rules *at that face considered in isolation.*

3 The Schützenberger involution

The composition

$$\psi_R \circ \psi_B^{-1}\colon \mathrm{CPP}(\pi) \to \mathrm{CPP}(\pi) \tag{7}$$

is well-defined: take a cylindric plane partition, decompose it via the Burge bijection into a pair $(\gamma, \mathfrak{d})$ with $\gamma \in \mathrm{IP}$ and $\mathfrak{d} \in \mathrm{ALCD}(\pi)$, then reassemble via the RSK bijection. This preserves the base partition γ and the face labels (the ALCD is unchanged). In the standard (non-Q-enriched) case, this is a weight-preserving endomorphism on $\mathrm{CPP}(\pi)$.

Remark 2. From [3]: "Two natural local rules exist (RSK and Burge), related by the Schützenberger involution. Their composition $\psi_R \circ \psi_B^{-1}\colon \mathrm{CPP}(\pi) \to \mathrm{CPP}(\pi)$ has a conjectural interpretation via crystal bases for $\widehat{\mathfrak{sl}}_n$."

4 The propagation obstruction

Here is the critical failure. Consider a cylindric growth diagram with multiple rows of faces.

4.1 First row

The inputs (ν, α, β) at each first-row face are determined by the base partition γ. These are the same for both Burge and RSK. Therefore:

- Strip sizes at first-row faces: **identical** for both rules.
- Q-weights at first-row faces: **identical**.
- Output partition λ: **different** ($\lambda_B \neq \lambda_R$ in general).

4.2 Second row and beyond

At the second row, the inputs include λ from the first row. Since $\lambda_B \neq \lambda_R$, the strip sizes a', b' at second-row faces are **different** between Burge and RSK. Therefore:

$$Q_{a'+k} \cdot Q_{b'+k} \quad \text{differs between the two rules.}$$

Proposition 3 (Propagation obstruction)**.** *The Q-weights propagate upward through the growth diagram. The Schützenberger involution changes the partition shapes at interior vertices, which changes the strip sizes at subsequent faces, which changes the Q-weights. The total Q-weight (the product over all faces) is* ***not*** *preserved.*

Schematically:

Standard case	**Q-enriched case**
Face weight $= 1$ (always)	Face weight $= Q_{a+k} \cdot Q_{b+k}$
Independent of strip sizes	Depends on strip sizes
Schützenberger preserves total weight trivially	Schützenberger changes total Q-weight via propagation

5 The $d = 1$ exception

When $d = 1$, $Q_n = 1$ for all n. The Q-weight $Q_{a+k} \cdot Q_{b+k} = 1$ regardless of a, b, k. The strip-size dependence vanishes, and the Schützenberger involution trivially preserves Q-weights. This recovers the standard Burge–RSK equivalence.

6 Context-sensitivity and global coupling

The failure is fundamentally because the Q-kernel is *context-sensitive*: $\Phi_{a,b}(t)$ depends on the strip sizes (a, b), which are determined by the global partition structure. In the standard case, the face-label generating function is $1/(1-t)$ everywhere (universal), so any rearrangement of the growth diagram preserves weights. The Q-enrichment breaks this universality.

Remark 4 (Why positivity is hard)**.** This explains, at a structural level, why the positivity proof for $Q_{n,c}(q)$ is hard: any bijective technique that rearranges the growth diagram *globally* (Schützenberger, evacuation, jeu de taquin) will change the Q-weights unless it preserves the exact strip sizes at every face. The Q-kernel creates a **global coupling** between the local rule and the partition geometry.

7 What could work instead

The failure of the global Burge–RSK approach suggests three alternative strategies, all based on *local* operations.

7.1 Peel/glue at the ALCD level

The sign-reversing involution from [4] operates on pairs $(S, \mathfrak{d})$ where $\mathfrak{d}$ is an ALCD. The peel/glue mechanism uses Burge forward/inverse on a *single* face (the outermost diagonal), which does *not* change the rest of the growth diagram. This is local and self-consistent. The Q-weight change at the peeled face is controlled:

peel: removes one face with weight $Q_{a+k} \cdot Q_{b+k}$,
glue: adds one face with weight $Q_{a+k} \cdot Q_{b+k}$.

The rest of the growth diagram is untouched, so its Q-weights are preserved. The involution pairs:

$$(S,\, \mathfrak{d}) \;\longleftrightarrow\; (S \triangle \{k\},\, \mathfrak{d}'),$$

where $\mathfrak{d}'$ differs from $\mathfrak{d}$ by one face on the outermost diagonal.

7.2 The Q-kernel recurrence

The shifted factorisation ([1], Part III, Proposition 16.4):

$$\Phi_{a,b}(t) = Q_a\, Q_b + t\, \Phi_{a+1,b+1}(t) \tag{8}$$

peels off one layer of Q-weight at a time. An involution that peels/glues one diagonal at a time tracks the Q-weight shift through this recurrence. The $k = 0$ term $Q_a Q_b$ is the "uncoupled" contribution, and the remainder $t\, \Phi_{a+1,b+1}(t)$ represents the cross-level interactions that the involution must cancel.

Iterating (8):

$$\Phi_{a,b}(t) = \sum_{j=0}^{N-1} Q_{a+j}\, Q_{b+j}\, t^j + t^N\, \Phi_{a+N,b+N}(t). \tag{9}$$

So the involution can work layer by layer, cancelling one cross-level term at a time. This matches the Franklin-style structure where each involution step moves exactly one element between S and the cylindric object.

7.3 The commutator approach

From [1], Part III, the commutator isolates the $k \geq 1$ terms:

$$[\mathbf{Q}(z),\, \mathbf{Q}^*(w)] = -\sum_{a,b\geq 0} P_a\, T_b\, z^a w^b\, \big(\Phi_{a,b}(zw) - Q_a\, Q_b\big). \tag{10}$$

The obstruction to commutativity is precisely:

$$\Phi_{a,b}(t) - Q_a\, Q_b = t\, \Phi_{a+1,b+1}(t) = \sum_{k\geq 1} Q_{a+k}\, Q_{b+k}\, t^k. \tag{11}$$

A successful involution must cancel precisely these cross-level terms. The peel/glue mechanism applied to the outermost face of the ALCD amounts to cancelling the $k = 1$ term of this sum, then iterating inward.

8 The structural lesson

The Burge $\leftrightarrow$ RSK involution is a *global* transformation: it changes all internal vertices of the growth diagram simultaneously. But the Q-enrichment makes weights *locally coupled* to the partition shapes at those vertices.

Theorem 5 (Locality principle)**.** *Effective involutions for Q-positivity must be **local**—operating on one diagonal/face at a time—because the Q-kernel's shifted factorisation*

$$\Phi_{a,b}(t) = Q_a\, Q_b + t\, \Phi_{a+1,b+1}(t)$$

is itself a one-layer-at-a-time recurrence.

This is the algebraic reason why the Franklin-style peel/glue approach ([4]) is the right framework: it matches the recursive structure of the Q-kernel. The involution peels one face at a time, each step corresponding to one application of the shifted factorisation, and the Q-weights are tracked through the recurrence rather than through a global diagram transformation.

9 Open question

Can the local peel/glue involution be made explicit?

Conjecture 6 (Canonical peel)**.** *Given $(S, \mathfrak{d})$ with $\mathfrak{d} \in \mathrm{ALCD}_{n-j}(c)$, there exists a canonical choice of face on the outermost diagonal to peel, satisfying:*

1. Involutivity. $\mathrm{peel}(\mathrm{glue}(S, \mathfrak{d})) = (S, \mathfrak{d})$ *and vice versa.*
2. Weight preservation. $\sigma(S') + |\mathfrak{d}'| = \sigma(S) + |\mathfrak{d}|$.
3. Fixed points. *The unpaired elements at $j = 0$ contribute exactly $Q_{n,c}(q)$.*

The cyclic descent injection ([5], Section 8) provides a partial answer for the max-1 CPP case ($d \not\equiv 0 \pmod 3$), but the general case—where the ALCD has arbitrary depth—remains open. The Q-kernel framework suggests that the canonical choice should be guided by the shifted factorisation: peel the face whose removal corresponds to the $k = 1 \to k = 0$ reduction in $\Phi_{a,b}(t)$.

10 The pair-based Q-local rule

The preceding sections showed that naïvely lifting the Schützenberger involution to the Q-enriched setting fails because of the propagation obstruction: partition diamonds couple Q-weights to vertex shapes. We now take a fundamentally different approach. The underlying combinatorial object for the Q-commutator is **not** a partition diamond, but a **pair of ground-set elements**.

10.1 Why not diamonds

The standard Burge/RSK local rule operates on a diamond $(\nu;\, \alpha, \beta) \mapsto \lambda$ of integer partitions. This is the bijective incarnation of $T_n P_m = \sum_k P_{m-k} T_{n-k}$, which acts on partitions (the Schur basis of Λ).

The Q-operator $\mathbf{Q}(z) = \sum Q_m z^m P_m$ packages the cylindric Kostka polynomials as *coefficients* of the Pieri operators. Each Q_m is itself the signed generating function of the ground set $\mathcal{G}_{m,c}$:

$$Q_m = \sum_{(S,\mathfrak{d}) \in \mathcal{G}_{m,c}} (-1)^{|S|}\, q^{\sigma(S)+|\mathfrak{d}|}. \tag{12}$$

The cascade $T_n P_m = \sum_k P_{m-k} T_{n-k}$ still happens *inside*, but the Q-weights are *external* to the cascade—they come from the ground-set decomposition. A local rule on diamonds captures the cascade but misses the cylindric structure that Q_m encodes.

10.2 The underlying object: pairs of ground-set elements

The product $\mathbf{Q}^*(w)\, \mathbf{Q}(z)$ involves terms $Q_m\, Q_n\, z^m w^n\, T_n P_m$ (cf. [1], Part II). At the combinatorial level, the factor $Q_m \cdot Q_n$ is the signed generating function of the **product ground set**:

$$Q_m \cdot Q_n = \sum_{\substack{(S,\mathfrak{d}) \in \mathcal{G}_{m,c} \\ (S',\mathfrak{d}') \in \mathcal{G}_{n,c}}} (-1)^{|S|+|S'|}\, q^{\sigma(S)+|\mathfrak{d}|+\sigma(S')+|\mathfrak{d}'|}. \tag{13}$$

After the cascade with $m = a+k$ and $n = b+k$, the coefficient of $P_a\, T_b\, z^a w^b\, (zw)^k$ in $\mathbf{Q}^*(w)\, \mathbf{Q}(z)$ is $Q_{a+k}\, Q_{b+k}$.

Definition 7 (Pair-based Q-local rule)**.** The underlying combinatorial object at cascade level k with strip sizes (a, b) is a *pair of ground-set elements*:

$$\big((S, \mathfrak{d}),\ (S', \mathfrak{d}')\big)\ \in\ \mathcal{G}_{a+k,c} \times \mathcal{G}_{b+k,c}, \tag{14}$$

with:

- **Sign:** $(-1)^{|S|+|S'|}$.
- **q-weight:** $q^{\sigma(S)+|\mathfrak{d}|+\sigma(S')+|\mathfrak{d}'|}$.
- **Cascade parameter:** $k \geq 0$.

The Q-kernel is then expressed as a signed sum over pairs:

$$\Phi_{a,b}(t) = \sum_{k\geq 0} t^k \sum_{\substack{(S,\mathfrak{d})\in\mathcal{G}_{a+k,c}\\ (S',\mathfrak{d}')\in\mathcal{G}_{b+k,c}}} (-1)^{|S|+|S'|}\, q^{\sigma(S)+|\mathfrak{d}|+\sigma(S')+|\mathfrak{d}'|}. \tag{15}$$

10.3 The sign structure

The sign $(-1)^{|S|+|S'|}$ depends on the *total subset size* $|S|+|S'|$, which ranges from 0 to $(a+k)+(b+k)$. The terms decompose as:

$\lvert S\rvert + \lvert S'\rvert$	**Sign**	**Contribution**
0 (both empty)	$+1$	$\sum_{\mathfrak{d},\mathfrak{d}'} q^{\lvert\mathfrak{d}\rvert+\lvert\mathfrak{d}'\rvert}$
1 (one singleton)	-1	cancels against $\lvert S\rvert + \lvert S'\rvert = 0$ terms
2	$+1$	cancels against $\lvert S\rvert + \lvert S'\rvert = 1$ terms
$\vdots$	$\vdots$	$\vdots$

Remark 8 (Two sources of sign)**.** The signs arise from two *independent* copies of the alternating sum (12): one from $\mathcal{G}_{a+k}$ (the "creation" side) and one from $\mathcal{G}_{b+k}$ (the "annihilation" side). If each factor Q_{a+k} and Q_{b+k} were independently non-negative, the product would be non-negative. The pair structure opens the possibility of a *coupled* involution that works even when the individual factors are hard to handle.

10.4 The cascade as coupling

The cascade parameter k **couples** the two components of the pair: the first component lives in $\mathcal{G}_{a+k,c}$ and the second in $\mathcal{G}_{b+k,c}$, with k shared. This coupling is absent in the standard Burge rule (where the face label k carries weight 1 regardless of context) and is the essential new feature of the pair-based framework.

An involution on pairs can exploit the shared cascade in ways that two independent involutions cannot:

1. **Cross-pair exchange.** Move an element from S to S' (or vice versa), adjusting the ALCD components to preserve total weight. This changes the parity of $|S| + |S'|$ by 0 (not useful for sign cancellation) but can redistribute weight between the two components.

2. **Cascade shift.** Move from cascade level k to level $k\pm1$ by transferring data between the pair and the cascade. This changes the ground-set levels from $(a+k, b+k)$ to $(a+k\pm1, b+k\pm1)$ and modifies the pair accordingly.

3. **Coupled peel/glue.** In the first component, peel an element from $\mathfrak{d}$ and deposit it in S (changing the sign); simultaneously, in the second component, glue an element from S' onto $\mathfrak{d}'$ (changing the sign back). The net sign change is $(-1)^{+1} \cdot (-1)^{-1} = 1$ (no sign change), but the *weight* can be redistributed.

4. **Asymmetric peel.** Peel in only *one* component: add an element to S (sign changes by -1), adjusting the weight. This is the analogue of the Franklin-style involution, but applied to one component of the pair.

Remark 9 (The cascade recurrence, revisited)**.** The shifted factorisation $\Phi_{a,b}(t) = Q_a Q_b + t\,\Phi_{a+1,b+1}(t)$ acquires a new interpretation in the pair framework. The $k = 0$ term $Q_a Q_b$ corresponds to pairs in $\mathcal{G}_{a,c} \times \mathcal{G}_{b,c}$ (no cascade). The remainder $t\,\Phi_{a+1,b+1}(t)$ corresponds to pairs in $\mathcal{G}_{a+1,c} \times \mathcal{G}_{b+1,c}$ and higher (with cascade $k \geq 1$). The involution must map between these two regimes, using the cascade as the transfer mechanism.

11 The involution programme on pairs

11.1 The full signed sum

Assembling all the pieces, the Q-kernel at cascade level k is a signed sum over pairs:

$$[t^k]\,\Phi_{a,b}(t) = Q_{a+k}\,Q_{b+k} = \sum_{\substack{(S,\mathfrak{d})\in\mathcal{G}_{a+k}\\(S',\mathfrak{d}')\in\mathcal{G}_{b+k}}} (-1)^{|S|+|S'|}\, q^{\sigma(S)+|\mathfrak{d}|+\sigma(S')+|\mathfrak{d}'|}.$$

We need a sign-reversing, weight-preserving involution ι on the set

$$\mathcal{P}_{a,b,k} := \mathcal{G}_{a+k,c} \times \mathcal{G}_{b+k,c} \tag{16}$$

whose fixed points have $|S| + |S'|$ even and contribute $Q_{a+k}\,Q_{b+k} \in \mathbb{Z}_{\geq 0}[q]$.

11.2 Factored vs. coupled involutions

The simplest approach is a *factored involution*: apply a sign-reversing involution ι_1 on $\mathcal{G}_{a+k}$ independently of a sign-reversing involution ι_2 on $\mathcal{G}_{b+k}$, and set $\iota = \iota_1 \times \iota_2$.

Proposition 10. *If sign-reversing involutions ι_1 on $\mathcal{G}_{a+k,c}$ and ι_2 on $\mathcal{G}_{b+k,c}$ exist (proving $Q_{a+k} \geq 0$ and $Q_{b+k} \geq 0$ individually), then $\iota = \iota_1 \times \iota_2$ is a sign-reversing involution on $\mathcal{P}_{a,b,k}$.*

Proof. If $(S, \mathfrak{d})$ is paired with $(T, \mathfrak{e})$ by ι_1 (so $|S| \not\equiv |T| \pmod 2$), then for any $(S', \mathfrak{d}')$, the pairs $((S, \mathfrak{d}), (S', \mathfrak{d}'))$ and $((T, \mathfrak{e}), (S', \mathfrak{d}'))$ have opposite signs and the same weight. The product involution pairs them. □

This reduces the pair problem to the original single-factor problem and is therefore no easier. The interest of the pair framework lies in the possibility of a **coupled involution** that uses the cascade to transfer data between components.

11.3 Coupled involutions via cascade shift

Definition 11 (Cascade-coupled involution)**.** A *cascade-coupled involution* on $\bigsqcup_{k\geq 0} \mathcal{P}_{a,b,k}$ is a sign-reversing, weight-preserving involution ι that may pair elements at *different* cascade levels:

$$\iota\colon \big((S,\mathfrak{d}),\,(S',\mathfrak{d}'),\,k\big) \;\longmapsto\; \big((T,\mathfrak{e}),\,(T',\mathfrak{e}'),\,k'\big),$$

where $k' \neq k$ is allowed, provided that:

1. $|S| + |S'| \not\equiv |T| + |T'| \pmod 2$ (sign reversal),
2. $\sigma(S) + |\mathfrak{d}| + \sigma(S') + |\mathfrak{d}'| + k = \sigma(T) + |\mathfrak{e}| + \sigma(T') + |\mathfrak{e}'| + k'$ (weight preservation, where the cascade contributes weight k via the t^k monomial),
3. $\iota^2 = \mathrm{id}$ (involutivity).

The cascade shift is the key new feature: an element at cascade level k with pair data $((S,\mathfrak{d}),(S',\mathfrak{d}'))$ can be paired with an element at level k' by transferring weight between the cascade parameter and the pair components.

Remark 12 (The shifted factorisation as cascade shift)**.** The shifted factorisation $\Phi_{a,b}(t) = Q_a Q_b + t\,\Phi_{a+1,b+1}(t)$ says: the $k=0$ pairs in $\mathcal{G}_a \times \mathcal{G}_b$ and the $k \geq 1$ pairs in $\bigsqcup_{k\geq 1} \mathcal{G}_{a+k} \times \mathcal{G}_{b+k}$ must combine to give non-negative coefficients. A cascade-coupled involution can pair an element at level $k=0$ with an element at level $k=1$ (or higher), using the cascade parameter to absorb the weight difference.

11.4 The programme

1. **Define the pair set.** $\mathcal{P}_{a,b} := \bigsqcup_{k\geq 0} \mathcal{G}_{a+k,c} \times \mathcal{G}_{b+k,c}$, with sign $(-1)^{|S|+|S'|}$ and weight $\sigma(S) + |\mathfrak{d}| + \sigma(S') + |\mathfrak{d}'| + k$.
2. **Construct a cascade-coupled involution** ι on $\mathcal{P}_{a,b}$. The involution may pair elements at different cascade levels, using the cascade parameter as a "bridge" between the two ground-set components.
3. **Identify the fixed points.** The unpaired elements of ι contribute $\Phi_{a,b}(t) \in \mathbb{Z}_{\geq 0}[q][\![t]\!]$, which implies $Q_{a+k}Q_{b+k} \in \mathbb{Z}_{\geq 0}[q]$ for each k.
4. **Deduce single-factor positivity.** Setting $b=0$ gives $\Phi_{a,0}(t) = C_a(t) = \sum_k Q_{a+k}Q_k\,t^k$. Since $Q_0 = 1$, the $k=0$ term is Q_a, and non-negativity of $\Phi_{a,0}(t)$ implies $Q_a \geq 0$.

Remark 13 (Advantage over the factored approach)**.** Step (4) is the key advantage of the pair framework: proving $\Phi_{a,0}(t) \geq 0$ for all a is *a priori* a different problem from proving $Q_n \geq 0$ for all n, and the pair structure with cascade coupling provides additional tools (cross-component transfer, cascade shifts) that are not available when working with a single ground set. In particular, the involution on $\mathcal{G}_{a+k} \times \mathcal{G}_k$ can exploit the asymmetry between the two components (one at level $a+k$, one at level k) to construct pairings that have no analogue in the single-factor setting.

12 The Q-Burge and Q-RSK rules

The standard Burge and RSK local rules differ in a single respect: the cascade direction applied to overlapping columns in conjugate coordinates.

Rule	**Cascade direction**
Burge forward	UP: $\varepsilon(i) = \min\{j > i : j \notin A \cup B \cup C\}$
RSK forward	DOWN: $\delta(i) = \max\{j < i : j \notin A \cup B \cup C\}$

Both have the same type signature and weight condition; they produce different outputs only when the overlap $A \cap B \neq \emptyset$. We now lift this dichotomy to the pair framework of Section 10.

12.1 Peel and glue on ground-set elements

Let $(S, \mathfrak{d}) \in \mathcal{G}_{n,c}$ with $S \subseteq \{1, \ldots, n\}$, $|S| = j$, $\mathfrak{d} \in \mathrm{ALCD}_{n-j}(c)$. The *peel* and *glue* operations transfer data between the subset S and the ALCD $\mathfrak{d}$, using the local rule as the transfer mechanism.

Definition 14 (Peel)**.** Suppose $\mathfrak{d}$ has a non-trivial outermost face. The *R-peel* (for $R \in \{\mathrm{Burge}, \mathrm{RSK}\}$) applies the inverse local rule R^{-1} to extract the outermost face, producing:

$$(S, \mathfrak{d}) \xrightarrow{R\text{-peel}} (S \cup \{\ell\},\ \mathfrak{d}'),$$

where $\ell \in \{1, \ldots, n\} \setminus S$ is the position determined by the inverse rule (Burge^{-1} cascades **down**, RSK^{-1} cascades **up**), and $\mathfrak{d}' \in \mathrm{ALCD}_{n-j-1}(c)$ is the reduced ALCD. Weight is preserved: $\sigma(S \cup \{\ell\}) + |\mathfrak{d}'| = \sigma(S) + |\mathfrak{d}|$, and the sign changes: $(-1)^{j+1} = -(-1)^j$.

Definition 15 (Glue)**.** The inverse of peel. Given $i \in S$, the *R-glue* of i applies the forward local rule R to insert a horizontal strip of weight i into $\mathfrak{d}$:

$$(S, \mathfrak{d}) \xrightarrow{R\text{-glue of } i} (S \setminus \{i\},\ \tilde{\mathfrak{d}}),$$

with the same weight preservation and sign reversal.

In general, $\ell_{\mathrm{Burge}} \neq \ell_{\mathrm{RSK}}$ and $\mathfrak{d}'_{\mathrm{Burge}} \neq \mathfrak{d}'_{\mathrm{RSK}}$: the Burge inverse cascades the extracted data *downward* (tending to produce a smaller ℓ), while the RSK inverse cascades it *upward* (tending to produce a larger ℓ).

12.2 The Q-Burge rule on pairs

Definition 16 (Q-Burge rule)**.** On the pair set $\mathcal{P}_{a,b,k} = \mathcal{G}_{a+k,c} \times \mathcal{G}_{b+k,c}$, the *Q-Burge rule* $\mathfrak{B}$ applies a Burge-peel to the **first component**:

$$\mathfrak{B}\colon \big((S, \mathfrak{d}),\ (S', \mathfrak{d}')\big) \longmapsto \big((S \cup \{\ell_B\},\ \mathfrak{d}_B),\ (S', \mathfrak{d}')\big),$$

where ℓ_B and $\mathfrak{d}_B$ are determined by Burge^{-1} applied to the outermost face of $\mathfrak{d}$. The second component is unchanged.

12.3 The Q-RSK rule on pairs

Definition 17 (Q-RSK rule)**.** On the same pair set, the *Q-RSK rule* $\mathfrak{R}$ applies an RSK-peel to the **second component**:

$$\mathfrak{R}\colon \big((S,\mathfrak{d}),\ (S',\mathfrak{d}')\big) \longmapsto \big((S,\mathfrak{d}),\ (S'\cup\{\ell_R\},\ \mathfrak{d}'_R)\big),$$

where ℓ_R and $\mathfrak{d}'_R$ are determined by RSK^{-1} applied to the outermost face of $\mathfrak{d}'$. The first component is unchanged.

Both $\mathfrak{B}$ and $\mathfrak{R}$ are sign-reversing: each changes $|S|+|S'|$ by exactly 1, so the total sign $(-1)^{|S|+|S'|}$ flips.

Remark 18 (The double asymmetry)**.** The Q-Burge rule uses the Burge direction (cascade **down** for the inverse) on the *first* component, while the Q-RSK rule uses the RSK direction (cascade **up** for the inverse) on the *second* component. This double asymmetry—different cascade directions on different components of the pair—is the structural feature that makes the combined involution possible. It mirrors the standard Burge–RSK dichotomy, but applied to the two sides of the pair rather than to two bijections on the same object.

13 The paired involution

Having two sign-reversing operations on the pair set—one for each component, using opposite cascade directions—we combine them into a single sign-reversing involution.

13.1 The Franklin-style comparison

In the classical Franklin involution (for the pentagonal number theorem), one compares a *peel candidate* ℓ (the smallest part of the partition) with a *glue candidate* g (the end of the staircase): if $\ell < g$, one peels; if $g < \ell$, one glues; if $\ell = g$, the element is a fixed point.

In the pair framework, the two rules provide two peel candidates—one for each component—and the comparison between them determines which component to act on.

Definition 19 (The Q-Burge–RSK involution)**.** Let $p = ((S,\mathfrak{d}),\ (S',\mathfrak{d}'))$ be a pair in $\mathcal{P}_{a,b,k}$. Define the involution $\iota(p)$ as follows.

Step 1: Compute candidates.

- If $\mathfrak{d}$ is non-trivial, compute the Burge-peel candidate ℓ_B for the first component (cascade **down**: ℓ_B is the position where the extracted face data lands).
- If $\mathfrak{d}'$ is non-trivial, compute the RSK-peel candidate ℓ_R for the second component (cascade **up**: ℓ_R is the position where the extracted face data lands).

Step 2: Compare and act.

(P) **Peel phase.** If both candidates exist, compare ℓ_B and ℓ_R.

- If $\ell_B < \ell_R$: apply $\mathfrak{B}$ (peel first component).
- If $\ell_R < \ell_B$: apply $\mathfrak{R}$ (peel second component).
- If $\ell_B = \ell_R$: proceed to the glue phase.

If only one candidate exists, apply the corresponding rule.

(G) **Glue phase.** If the previous step produced a peeled pair p', this phase does not apply. Otherwise, check whether a glue operation can reverse a previous peel:

- If $S \neq \emptyset$ and $\max(S)$ is Burge-glueable (i.e., Burge-glue of $\max(S)$ followed by Burge-peel recovers $\max(S)$): apply Burge-glue to the first component.
- Else if $S' \neq \emptyset$ and $\min(S')$ is RSK-glueable: apply RSK-glue to the second component.

(F) **Fixed point.** If neither peel nor glue applies, p is a **fixed point** of ι.

The peel–glue duality ensures involutivity: if ι peels the first component of p to produce p', then ι applied to p' glues the first component back (since the newly added element $\ell_B \in S$ is the Burge-glue candidate, and gluing it reverses the peel).

13.2 Sign-reversal and weight preservation

Proposition 20. *The involution ι of Definition 19 is:*

1. ***Sign-reversing:*** *each non-fixed-point pair p satisfies $(-1)^{|S(\iota(p))|+|S'(\iota(p))|} = -(-1)^{|S(p)|+|S'(p)|}$, since exactly one of $|S|$, $|S'|$ changes by 1.*
2. ***Weight-preserving:*** *the total q-weight $\sigma(S) + |\mathfrak{d}| + \sigma(S') + |\mathfrak{d}'|$ is unchanged, since peel and glue individually preserve weight.*

13.3 Fixed-point structure

A pair $p = ((S, \mathfrak{d}), (S', \mathfrak{d}'))$ is a fixed point of ι when:

1. Both ALCDs $\mathfrak{d}$ and $\mathfrak{d}'$ are trivial (no face to peel), **and**
2. Both subsets S and S' are empty (no element to glue);

or when the peel candidates coincide ($\ell_B = \ell_R$) and no glue operation is available.

Remark 21 (The trivial fixed points)**.** In the first case, $|S| = |S'| = 0$, so the fixed point has sign $(-1)^0 = +1$ and contributes $q^{|\mathfrak{d}|+|\mathfrak{d}'|}$ with a *positive* coefficient. These trivial fixed points correspond to pure ALCD pairs $(\mathfrak{d}, \mathfrak{d}') \in \mathrm{ALCD}_{a+k}(c) \times \mathrm{ALCD}_{b+k}(c)$ and contribute the "uncancelled positive core" of $Q_{a+k} \cdot Q_{b+k}$.

13.4 Why two rules are needed

Remark 22 (Single rule vs. paired rules)**.** A single-rule involution (say, Burge-peel/glue on the first component alone) is a *factored involution* (Proposition 10): it proves $Q_{a+k} \geq 0$ for the first component independently, which is the original unsolved problem.

The paired involution uses *both* rules simultaneously, one per component, with opposite cascade directions. The cross-component comparison in Step 2(P) creates pairings that have no analogue in the single-factor setting:

- A pair where the first component is a fixed point of Burge-peel may still be paired by the Q-RSK rule acting on the second component.
- The opposite cascade directions (Burge **down** on first, RSK **up** on second) ensure that the peel candidates ℓ_B and ℓ_R explore different parts of the position space, maximising the chance that at least one operation is available.

13.5 Connection to the shifted factorisation

The shifted factorisation $\Phi_{a,b}(t) = Q_a Q_b + t\,\Phi_{a+1,b+1}(t)$ acquires a concrete meaning in this framework. The $k = 0$ pairs $\mathcal{G}_a \times \mathcal{G}_b$ are acted on by the involution ι restricted to cascade level 0. The unpaired (fixed-point) pairs at level 0 contribute $Q_a Q_b$ to $\Phi_{a,b}(0)$. At level $k = 1$, the involution acts on $\mathcal{G}_{a+1} \times \mathcal{G}_{b+1}$, and the fixed-point contribution is $Q_{a+1}Q_{b+1}$, the coefficient of t. In general:

$$[t^k]\,\Phi_{a,b}(t) = \sum_{\substack{p \in \mathcal{P}_{a,b,k} \\ \iota(p) = p}} q^{\mathrm{wt}(p)} \;\in\; \mathbb{Z}_{\geq 0}[q],$$

provided ι has only positive fixed points at each cascade level. Setting $b = 0$ recovers Q_a as the $k = 0$ fixed-point contribution of the involution on $\mathcal{G}_a \times \mathcal{G}_0$, where $\mathcal{G}_0 = \{(\emptyset, \mathfrak{d}_0)\}$ is a singleton (the trivial ALCD at level 0).

References

[1] R. Langer, *The Q-Operator and its Adjoint: From Level 3 Operator Traces to Cascade Commutators*, companion note, 2026.

[2] A. Borodin, *Periodic Schur process and cylindric partitions*, Duke Math. J. **140** (2007), 391–468.

[3] R. Langer, *From Commutation Relations to Growth Diagrams: A Summary of the Operator–Bijection Correspondence*, companion note.

[4] R. Langer, *Towards an Involution Proof of Positivity for $Q_{n,c}(q)$*, companion note.

[5] R. Langer, *Weighted Local Rules and the Cyclic Descent Injection*, companion note.

[6] R. Langer, *The Underlying Set: Cylindric Plane Partitions, ALCDs, and the Ground Set for the Involution*, companion note.

[7] R. Langer, *Cylindric Partitions and Bijective Proofs*, PhD thesis, University of Melbourne.

[8] M. A. A. van Leeuwen, *The Robinson–Schensted and Schützenberger algorithms, an elementary approach*, Electron. J. Combin. **3** (1996), R15.